

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

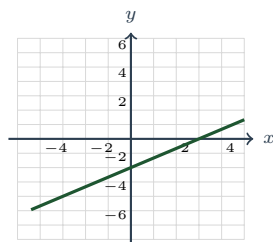
Write the letter of the best description next to each term.

- | | |
|-----------------------------|--|
| 1. Counterexample _____ | (A) The number $b^2 - 4ac$, which counts a quadratic's real solutions <i>before</i> you solve it. |
| 2. Closure _____ | (B) An equation true for every real number, so its solution set is all real numbers. |
| 3. Factor completely _____ | (C) One example that proves a general claim false; no number of successes can replace it. |
| 4. Literal equation _____ | (D) The property that performing an operation on members of a set always lands you back inside that set. |
| 5. Identity _____ | (E) An equation with more than one letter, solved for one letter in terms of the others. |
| 6. Discriminant _____ | (F) To rewrite an expression as a product — GCF first — until no factor breaks down further over the integers. |
| 7. Zero of a function _____ | (G) The constant ratio of change in output to change in input for a linear function; the slope. |
| 8. Rate of change _____ | (H) An input whose output is 0; also a solution of $f(x) = 0$ and an x -intercept of the graph. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- Which property justifies the step $5(x - 3) = 5x - 15$?
 (A) Commutative property of multiplication (B) Distributive property
 (C) Associative property of multiplication (D) Multiplicative identity
- Simplified with no negative exponents, $\frac{18a^5b^{-3}}{6a^2b}$ is
 (A) $\frac{3a^3}{b^4}$ (B) $\frac{3a^3}{b^2}$ (C) $3a^3b^4$ (D) $\frac{12a^3}{b^4}$
- The solution of $-4x + 9 \geq 25$ is
 (A) $x \geq -4$ (B) $x \leq -4$ (C) $x \leq 4$ (D) $x \geq 4$
- How many *real* solutions does $2x^2 - 3x + 5 = 0$ have?
 (A) two (B) one (C) none (D) infinitely many
- Which relation is *not* a function?
 (A) $\{(-3, 1), (0, 1), (2, 1), (5, 1)\}$ (B) $\{(1, -3), (1, 0), (2, 5)\}$
 (C) $\{(-2, 4), (-1, 1), (0, 0), (1, 1)\}$ (D) $y = 3x - 7$
- SOL-style.** Which equation represents the line shown?



(A) $y = \frac{3}{2}x - 2$

(B) $y = \frac{2}{3}x - 2$

(C) $y = -\frac{2}{3}x - 2$

(D) $y = \frac{2}{3}x + 2$

Part C — Short Answer & Computation (40 pts)

Show your work. A wrong answer with a right idea can still earn credit.

1. Numbers and justification.

(a) Label each number **R** (rational) or **I** (irrational).

$\sqrt{49}$ _____ $\sqrt{7}$ _____ $-\frac{5}{8}$ _____ $0.\overline{3}$ _____ π _____

(b) Name the property that justifies each step.

$2(x + 5) = 2x + 10$ _____

$2x + 10 - 10 = 14 - 10$ _____

$3 + x = x + 3$ _____

(c) “The set of integers is closed under division.” Disprove it.

2. Translate and evaluate.

(a) Write an expression for “seven less than four times a number n .” _____

(b) Evaluate $\frac{|a - b|}{2} + \sqrt{c}$ for $a = -3$, $b = 5$, $c = 16$.

(c) Evaluate $\sqrt[3]{x} - 2y$ for $x = -27$, $y = \frac{1}{2}$.

3. Exponents and polynomial operations.

(a) $(3x^2y^4)(-2x^5y)$ (b) $\frac{20p^6q^2}{5p^6q^5}$ (no negative exponents)

(c) $(2x - 5)(3x + 4)$ (d) $(4x^2 - x + 6) - (x^2 + 3x - 2)$

4. Factor completely. GCF first, every time.

(a) $3x^2 - 27$

(b) $x^2 - 2x - 3$

(c) $2x^2 + 11x + 12$

(d) $4x^3 - 12x^2 + 8x$

5. **Radicals and rational exponents.**

(a) $\sqrt{50} =$ _____ (b) $\sqrt[3]{40} =$ _____

(c) Write $\sqrt{x^3}$ with a rational exponent. _____

(d) Write $8^{1/3}$ as a radical, then evaluate it. _____

(e) $\sqrt{12} + \sqrt{27} =$ _____

6. **Linear equations and inequalities.**

(a) Solve $3(2x - 4) = 4x + 2$.

(b) Classify each solution set as **one solution**, **no solution**, or **infinitely many**. You should not need to finish solving.

$5x + 3 = 5x - 8$ _____ $2(x + 6) = 2x + 12$ _____

(c) Solve $A = \frac{1}{2}bh$ for h . _____

(d) Solve $-2x + 1 < 9$, then graph the solution on the number line.

x _____



7. **Quadratic equations.**

(a) Solve by factoring: $x^2 - 2x - 3 = 0$

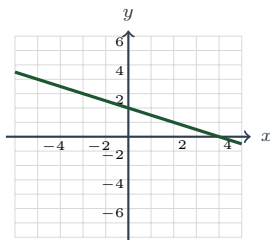
(b) Solve by square roots: $(x + 1)^2 = 49$

(c) Solve with the quadratic formula: $x^2 + 4x - 6 = 0$ (exact answer)

(d) *Without solving*, use the discriminant to say how many real solutions $9x^2 - 12x + 4 = 0$ has. _____

Discriminant: _____

8. **Read the graph.** The linear function g is graphed below; each gridline is one unit.



(a) $g(-2) =$ _____ (b) Solve $g(x) = 0$: $x =$ _____

(c) That number in (b) is called the _____ of g , and on the graph it is the point _____.

(d) Slope: _____; y -intercept: _____; equation of g : _____

(e) Is g increasing or decreasing? _____ Domain: _____

(f) Write the equation of the line through $(0, -3)$ **parallel** to g _____, and the line through $(0, 2)$ **perpendicular** to g _____.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **Three families, six questions.** Each table below is generated by a linear, a quadratic, or an exponential rule.

x	$A(x)$	x	$B(x)$	x	$C(x)$
0	5	0	3	0	1
1	8	1	6	1	2
2	11	2	12	2	5
3	14	3	24	3	10

(a) Name the family of each table and give the numerical evidence — constant difference, constant *second* difference, or constant ratio. (b) Write a rule for each function. (c) Which function is largest at $x = 1$? At $x = 5$? Show the values. (d) A classmate says “ B is just the biggest one.” Use your answers to (c) to explain why “which is bigger” is not a fair question until someone says *where*. (e) If B modeled a population of bacteria in hours, what domain would make sense, and why is that narrower than the domain of the rule by itself?

2. **Modeling and solution sets.** Two climbing gyms price a day pass differently.

Summit: a one-time \$40 membership fee, then \$12 per visit.

Basecamp: no fee, \$22 per visit.

(a) Write a function for the total cost at each gym after v visits. (b) Find the number of visits at which the two costs are equal. Show every step, and name the property of equality that justifies each one. (c) Verify your answer by substituting it into *both* functions. (d) Which gym is cheaper for 2 visits? For 10 visits? Give the costs. (e) Suppose Basecamp changed its pricing to a \$40 fee plus \$12 per visit. Setting the two costs equal would give $12v + 40 = 12v + 40$. What is the solution set, what is that kind of equation called, and what does it mean about the two gyms? (f) What domain makes sense for v in this situation, and why?